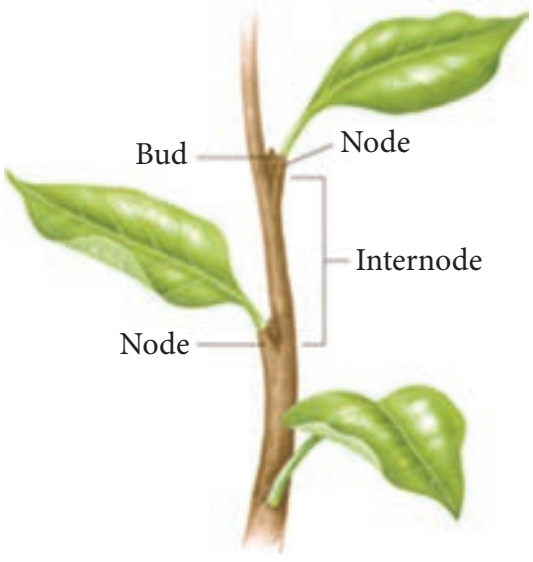
Stem

Stem grows above the soil, and it grows towards the sunlight. It has nodes and internodes. **Nodes** are the parts of stem, where leaf arises. The part of the stem between two successive nodes is called **internode.** The bud at the tip of the stem is known as apical or **terminal bud**, and the buds at the axils of the leaves are called **axillary buds**.



Activity 2

Conduction of water

Aim: To observe conduction of water by stem.

What you need? A small twig of balsam plant, a glass of water and a few drops of red ink.

What to do? Place the small twig in the water with red ink.

What do you see? The stem becomes reddish.

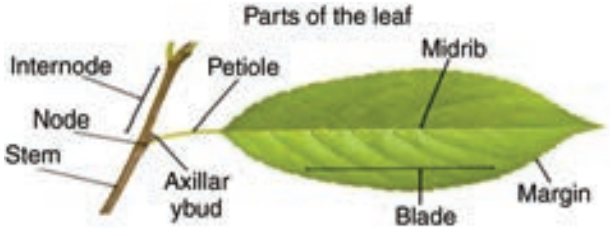
What do you learn? This is because red coloured water is being absorbed by the stem upwards.

Functions of the stem

- Supports the branches, leaves, flowers and fruits.
- Transports water and minerals from roots to upper aerial parts of the plant.
- Transports the prepared food from leaves to other parts through stem.
- Stores food as in the case of sugarcane.

Leaf

The leaf is a green, flat expanded structure borne on the stem at the node.



A leaf has a stalk called **petiole**. The flat portion of the leaf is called **leaf lamina** or **leaf blade**. On the lamina, there is a main vein called **midrib**. Other veins are branched out from mid rib. The portion of the leaf connected in the nodal region of the stem is known as the **leaf base**. Leaves of some plants possess a pair of lateral outgrowth on the base, on either side of axillary bud. These are called **stipules**.

The green colour of the leaf is due to the presence of green coloured pigment called **chlorophyll**. On the lower side of the leaf there are tiny pores or openings known as **stomata**.

Functions of the Leaves

- The green leaves prepare food by photosynthesis.
- They help in respiration.
- They carry out transpiration.